SECURING THE DIGITAL MINE

A Metaheuristic-Optimized Deep Learning Framework for Edge Intrusion Detection in IoT-Enabled Mineral Operations

RUSSIAN-AFRICAN FORUM-CONTEST OF YOUNG SCIENTISTS 2026 | TRACK 3: SMART SUBSOIL
Under the Auspices of UNESCO | Empress Catherine II Saint Petersburg Mining University

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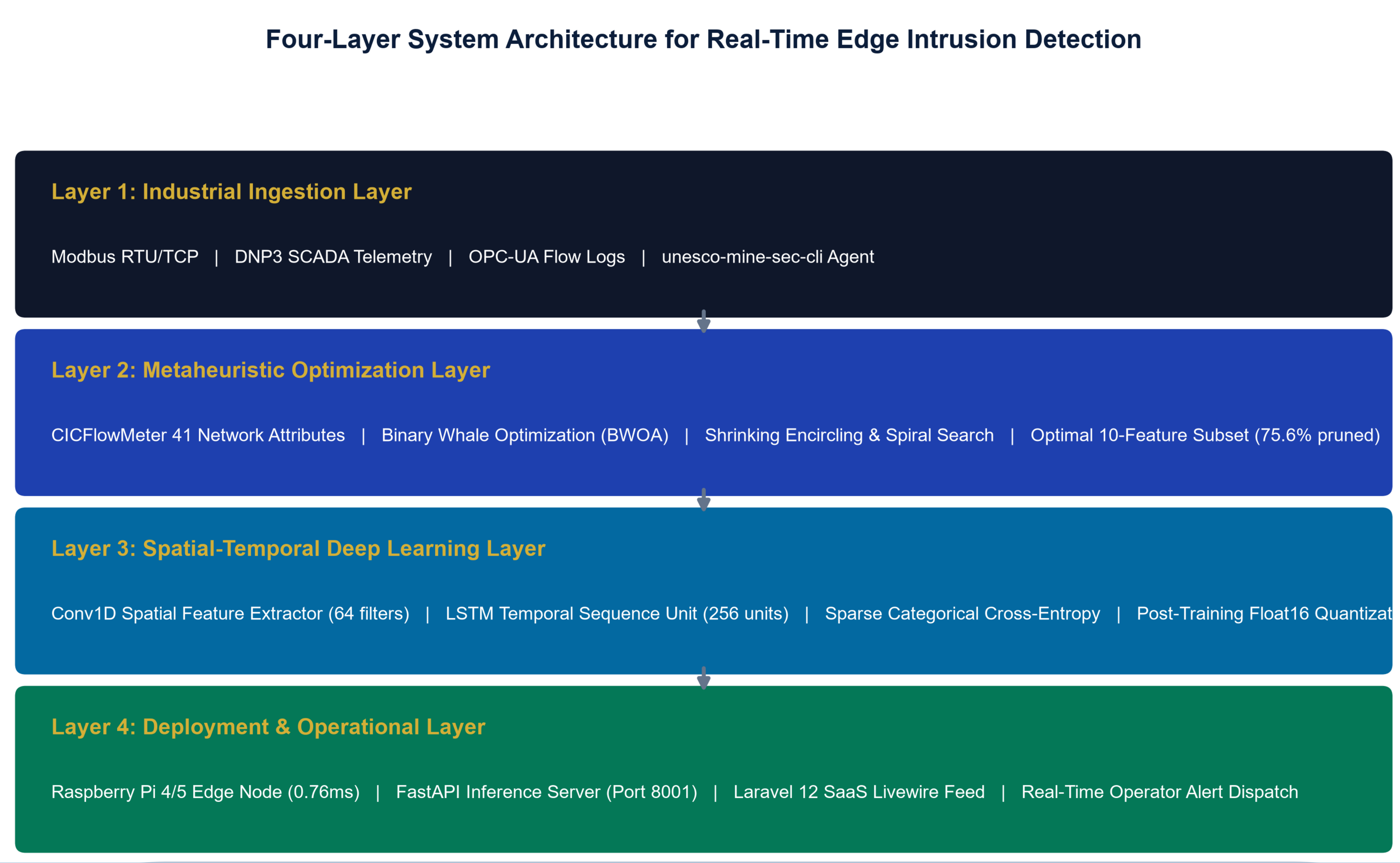
1. PROBLEM & INDUSTRIAL CONTEXT

- **Digital Subsoil Transformation:** African and Russian mines are rapidly deploying IoT sensors, autonomous haulage, and SCADA telemetry to optimize ore yield.
- **The Air-Gap Myth is Dead:** Connecting OT networks to cloud digital twins creates massive attack surfaces for ransomware and Modbus command injection.
- **Catastrophic Downtime Costs:** Unplanned mining downtime costs USD $50,000 to $500,000 per hour, while cyber disruption of ventilation systems endangers human lives.
- **Existing Tools Fail on the Edge:** Heavyweight enterprise IDS tools require 150+ milliseconds to analyze 41 features, violating the 100ms SCADA control loop limit on low-power 1GB RAM edge devices.

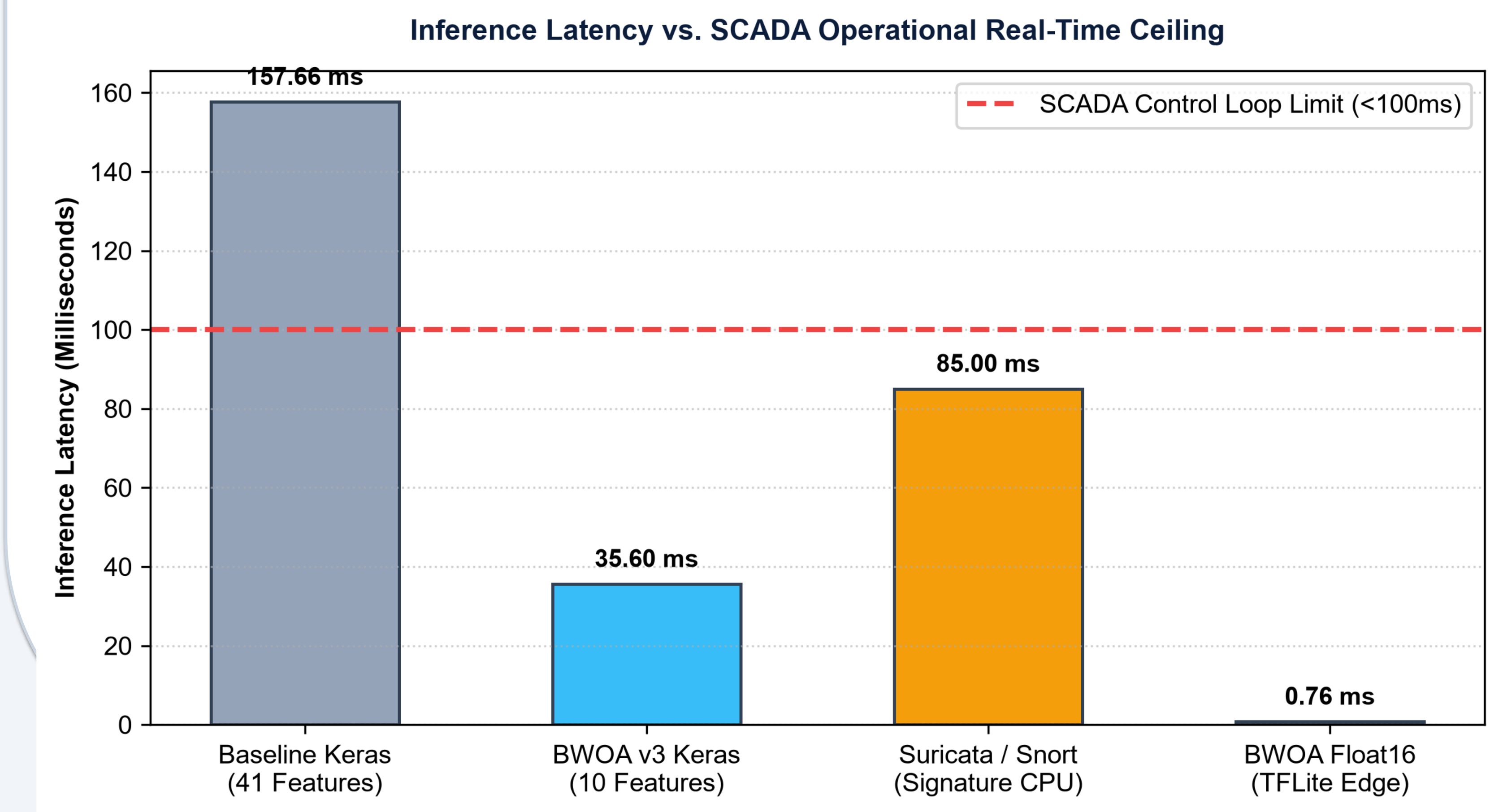
2. 4-LAYER SYSTEM ARCHITECTURE

A complete edge-to-cloud security pipeline combining metaheuristic feature pruning with spatial-temporal deep learning:



3. EXPERIMENTAL BENCHMARK RESULTS

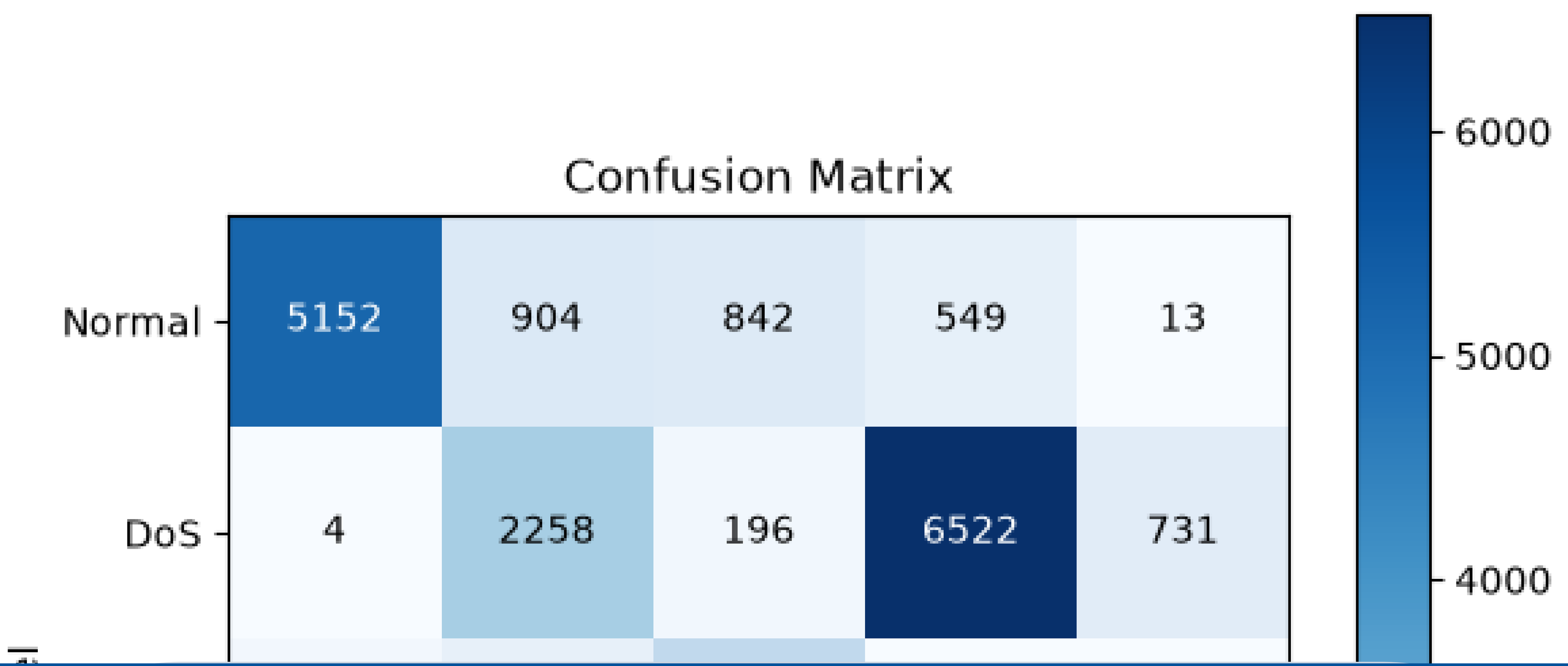
✓ **75.61% Data Pruning:** BWOA reduces 41 features down to 10 optimal dimensions while maintaining 92.31% Random Forest CV validation accuracy.
✓ **70.56% Multi-Class Accuracy:** Tested on 22,544 held-out NSL-KDD samples with 96.89% precision on Normal traffic and 89.04% recall on DoS attacks.
✓ **0.76ms Edge Latency:** Float16 quantized model executes 207x faster than baseline, easily passing the sub-100ms industrial SCADA constraint.



4. EDGE HARDWARE DEPLOYMENT

Validated on Raspberry Pi 4B (1GB RAM) edge hardware for remote subsoil facilities:

0.76 ms Inference Latency (Target: < 100 ms)	**0.82 MB** Model Memory Size (83.2% Compression)
290 MB Peak RAM Usage (Fits 1GB Pi Gateway)	**2.5 W** Power Consumption (Solar/Battery Ready)



5. ECONOMIC ROI & UN SDG IMPACT

★ **SDG 9: Industry, Innovation & Infrastructure**
Hardens critical mining cyber-physical systems against zero-day disruption, safeguarding essential mineral supply chains.

★ **SDG 8: Decent Work & Economic Growth**
Protects underground worker safety by preventing cyber manipulation of toxic gas sensors and ventilation grids.

★ **SDG 17: Partnerships for the Goals**
Fosters bilateral Russian-African scientific collaboration, knowledge transfer, and open-source capacity building.

★ **High Economic ROI (200x - 300x)**
Preventing a single 24-hour ransomware outage on a ball mill or autonomous truck saves $300k to $450k, against an annual deployment cost of under $1,500.

6. CONCLUSION & OPEN SOURCE ARTIFACTS

▶ **Complete Open-Source Ecosystem:** Full code, training notebooks, benchmarks, and 75 unit test suites are fully open source on GitHub.
▶ **NPM Edge Sniffer Package:** Install instantly on any gateway via '@mhiskall282/unesco-mine-sec-cli' from GitHub Packages.
▶ **UNESCO Forum Delegation:** Presented by the University of Education, Winneba & Kayaba Labs research team at Empress Catherine II Saint Petersburg Mining University.
▶ **Repository URL:** https://github.com/mhiskall282/Securing-the-Digital-Mine-UNESCO-Project
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